t3-w05 — condition C3

Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

> C2 plus a validation and repair pass over the draft.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	**present**

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

```
Mathematics
```

`grade_level`

```
JSS1
```

`topic`

```
Three dimensional shapes II
```

`subtopic`

```
Three dimensional figures: properties of cylinders and spheres, and volume
```

`term`

```
Third Term
```

`week`

```
5
```

`lesson_objectives`

- identify properties of cylinders and spheres
- find the volume of a cube and a cuboid

entry_behaviour

- properties of cubes and cuboid
- properties of pyramids and cones

instructional_materials

- shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

objective_text

identify properties of cylinders and spheres

current_step

1

instructional_events

- present_content
- elicit_performance
- present_content

student_tasks

- 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

source_figures


```
[
 {
  "caption": "A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "A sphere drawn as a circle with an ellipse across its middle to show the curved surface.",
  "figure_ref": "fig-02"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Properties of a cylinder\n\n**fig-01.** A cylinder drawn upright: an ellipse for the top face, two vertical sides\nfalling from its ends, and a curve closing the base.\n\n[drawing attached to this section as fig-01]\n\n1. It has two circular faces and one curved face\n2. It has no vertex\n3. It has two curved edges\n4. It has three faces altogether\n\n#### Properties of sphere\n\n**fig-02.** A sphere drawn as a circle with an ellipse across its middle to show the\ncurved surface.\n\n[drawing attached to this section as fig-02]\n\n1. A sphere is symmetrical from all direction\n2. It has no edge and no vertex\n3. It has one curved surface\n\n#### Student activity\n\n1. How many circular faces does a solid cylinder have?\n2. A cylinder has how many faces."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — find the volume of a cube and a cuboid

objective_text

find the volume of a cube and a cuboid

current_step

2

instructional_events

• present_content
• provide_learning_guidance

assessment_points

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

source_figures


```
[
 {
  "caption": "A triangular prism lying on its rectangular face. The triangular end has a base of
5 m and a height of 3 m, and the prism runs back 10 m.",
  "figure_ref": "fig-03"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Volume of cube and cuboid\n\n1. Volume of cuboid = Area of base $\\times$
height\n\n   $= \\text{length} \\times \\text{breadth} \\times \\text{height}$\n\n   $= lbh$
\n\n2. Volume of cube = Length $\\times$ Length $\\times$ Length\n\n   $= l^{3}$\n\n**Example
1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 3cm$\n\nbreadth $= 2cm$\n\nheight $=
8cm$\n\nVolume $= 3cm \\times 2cm \\times 8cm$\n\n$= 48cm^{3}$\n\n**Example 2.** The volume of a
cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the\nheight of the cuboid.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 6cm$\n\nbreadth $= 2cm$\n\nheight
$= ?$\n\nVolume $= 24cm^{3}$\n\n$24cm^{3} = 6cm \\times 2cm \\times h$\n\n$24cm^{3} = 12cm^{2}h$
\n\nDivide both side by 12cm$^{2}$\n\n$\\frac{24cm^{3}}{12cm^{2}} = \\frac{12cm^{2}h}{12cm^{2}}$
\n\n$2cm = h$\n\n$h = 2cm$\n\n#### Volume of cube\n\n**Example 3.** Calculate the volume of a
cube of side 3cm.\n\nSolution\n\nVolume of cube $= l^{3}$\n\n$V = (3cm)^{3}$\n\n$V = 3cm \\times
3cm \\times 3cm$\n\n$V = 27cm^{3}$\n\n#### Volume of prism\n\nCalculate the volume of the prism
below.\n\n**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a
base\nof 5 m and a height of 3 m, and the prism runs back 10 m.\n\n[drawing attached to this
section as fig-03, marked 5 m, 3 m, 10 m]\n\nSolution:\n\nArea of triangular face $= \\frac{1}{2}
bh$\n\n$= \\frac{1}{2} \\times 5 \\times 3$\n\n$= 7.5m^{2}$\n\nVolume of prism $=$ Area of base $
\\times$ height\n\n$= 7.5m^{2} \\times 10m$\n\n$= 75m^{3}$"
 }
]
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Properties of Cylinders and Spheres

> EXPLANATION
> A **cylinder** is a solid with two identical circular faces (the top and bottom) and one curved surface that connects them. It has:
> - 2 circular faces
> - 1 curved face
> - No vertices (points where edges meet)
> - 2 curved edges (the circles where the curved surface meets each face)
> - In total, 3 faces (the two circles plus the curved surface)
>
> A **sphere** is a perfectly round solid. Its surface is the same distance from the centre in every direction. It has:
> - No faces, edges, or vertices – the whole surface is one continuous curved surface
> - It is completely symmetrical from every direction
> - Only one curved surface
>
> These properties help you recognise each shape and describe them correctly.

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
>  \draw (-1.2,2.6) -- (-1.2,0);
>  \draw (1.2,2.6) -- (1.2,0);
>  \draw (-1.2,0) arc (180:360:1.2 and 0.35);
>  \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
> \end{tikzpicture}
> ```

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (0,0) circle (1.3);
>  \draw (-1.3,0) arc (180:360:1.3 and 0.42);
>  \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
> \end{tikzpicture}
> ```

1. How many circular faces does a solid cylinder have?
answer: 2

2. A cylinder has how many faces in total?
answer: 3

Objective 2

Finding the Volume of Cubes and Cuboids

EXPLANATION

The volume of a three-dimensional shape tells us how much space it occupies.

Cuboid – a box-shaped solid with length (l), breadth (b) and height (h). Its volume is found by multiplying the area of the base by the height:

$$V_{\text{cuboid}} = l \times b \times h$$

Cube – a special cuboid where all three edges are the same length (s). Its volume is therefore:

$$V_{\text{cube}} = s \times s \times s = s^{3}$$

Remember to keep the units consistent (all centimetres, all metres, etc.) and write the final answer with the unit cubed (cm^3, m^3).

WORKED_EXAMPLE

Example 1 – Volume of a cuboid

Find the volume of a cuboid whose length is 3 cm, breadth is 2 cm and height is 8 cm.

$$V = l \times b \times h = 3\,\text{cm} \times 2\,\text{cm} \times 8\,\text{cm} = 48\,\text{cm}^{3}$$

Example 2 – Finding a missing height

A cuboid has a volume of 24 cm^3 and its base measures 6 cm by 2 cm. What is its height?

$$V = l \times b \times h$$
$$24\,\text{cm}^{3} = 6\,\text{cm} \times 2\,\text{cm} \times h$$
$$24\,\text{cm}^{3} = 12\,\text{cm}^{2} \times h$$

Divide both sides by 12 cm^2:

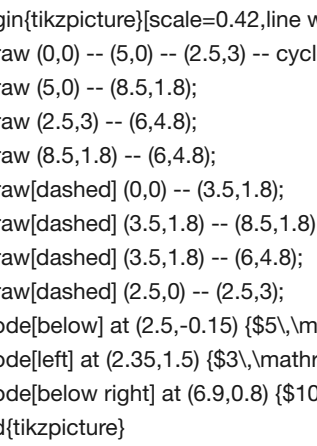
$$h = \frac{24\,\text{cm}^{3}}{12\,\text{cm}^{2}} = 2\,\text{cm}$$

Example 3 – Volume of a cube

Calculate the volume of a cube with side length 3 cm.

$$V = s^{3} = (3\,\text{cm})^{3} = 3\,\text{cm} \times 3\,\text{cm} \times 3\,\text{cm} = 27\,\text{cm}^{3}$$

VISUAL



1. 1. Find the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
answer: 96 cm^3

2. 2. Find the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
answer: 30 m^3

3. 3. What is the volume of a cube whose sides are each 4 cm?

answer: 64 cm³

4. 4. A rectangular tank holds 36 m³ of water and measures 6 m long by 2 m wide. How deep is the water?
answer: 3 m

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Enacted content	**grounded**
Worked example a problem with its solution worked through	1	Enacted content	**grounded**
Figure a drawing the learner reads from	3	Figures	**grounded**
Practice what the learner does	6	Set work	**grounded**
Answer what the learner checks against	6	Enacted content	**grounded**